

CS & IT

Discrete Mathematics

DPP: 2

Set Theory & Algebra

- Q1** Let A be $\{a, b, c\}$. Let the relation R on A and let $R = \{(b, a), (b, c), (c, a), (c, b), (b, b)\}$. Which of the following statements about R is/are true?
 (A) R is neither reflexive nor irreflexive.
 (B) R is not symmetric.
 (C) R is transitive.
 (D) R is not anti-symmetric.
- Q2** Let A and B be two sets such that $A \times B = \{(1, 1), (2, 2), (3, 1), (3, 2), (1, 2), (1, 4), (2, 1), (2, 4), (3, 4)\}$. What is the value of $|P(A)| + |P(B)|$? ____
- Q3** The relation R is defined on $Z = \{0, \pm 1, \pm 2, \pm 3 \dots\}$ by xRy if and only if $(x^2 + y^2) \bmod 4 = 0$. Then R is:
 (A) Reflexive, Symmetric
 (B) Transitive, Symmetric
 (C) Symmetric
 (D) Reflexive, Symmetric and Transitive
- Q4** Let R be the relation on the real numbers given by xRy iff $(x - y)^2 < 0$. Then which of the following statement is/are true?
 (A) R is an equivalence relation.
 (B) R is symmetric.
 (C) R is asymmetric and anti-symmetric both.
 (D) R is transitive.
- Q5** Let R be the relation on R (Where R is set of Natural Number) given by xRy if and only if $x < 2y + 1$, then R is ____
 (A) Reflexive, but not symmetric and not transitive
 (B) Reflexive, Symmetric, and not transitive
 (C) Not Reflexive, not symmetric and Transitive
 (D) Reflexive, but not symmetric and Transitive
- Q6** Let R be the relation on $A = \{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 1), (2, 3), (4, 4), (3, 4)\}$. What is the cardinality of the reflexive closure of R ? ____
- Q7** Let R be the relation on $A = \{1, 2, 3, 4, 5\}$ given by $R = \{(1, 2), (2, 2), (2, 4), (2, 3), (1, 1), (3, 3), (3, 2)\}$. What is the cardinality of the symmetric closure of R ? ____
- Q8** Let R be the relation on $A = \{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 3), (4, 1), (3, 4)\}$. What is the cardinality of the transitive closure of R ? ____
- Q9** Suppose that R is the relation on positive real numbers such that xRy if and only if $xy = 1$. Then R satisfies how many of the properties given below? ____
 (i) Reflexive
 (ii) Irreflexive
 (iii) Symmetric
 (iv) Anti-symmetric
 (v) Transitive
 (vi) Asymmetric
- Q10** Let R be the relation $\{(a, b) \mid a \neq b\}$ on the set of integers. What is the reflexive closure of R ?
 (A) $\{(a, b) \mid a = b\}$
 (B) $\{(a, b) \mid a \leq b\}$
 (C) $\{(a, b) \mid a \geq b\}$
 (D) $Z \times Z$

Answer Key


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Q1 (A, B, D)
Q2 16~16
Q3 (B)
Q4 (B, C, D)
Q5 (A)

Q6 8~8
Q7 9~9
Q8 16~16
Q9 1~1
Q10 (D)

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Hints & Solutions

Q1 Text Solution:
 $R = \{(b, a), (b, c), (c, a), (c, b), (b, b)\}$

(a) **True;** as $(a, a), (c, c)$ is not there,

$\Rightarrow$ Not reflexive

(b) **True;** as (a, b) is not there.

(c) **False;** as $(c, b), (b, c)$ is there but (c, c) is not there.

(d) **True;** due to existence of pair (b, c) & (c, b)

Q2 Text Solution:
 $A = \{1, 2, 3\}$ and $B = \{1, 2, 4\}$
 $|P(A)| = 2$
 $3 = 8$, and $|P(B)| = 2$
 $3 = 8$.

So, the value of $|P(A)| + |P(B)| = 8 + 8 = 16$.

Q3 Text Solution:

R is not reflexive because $(1, 1) \notin R$.

R is symmetric because if $(x^2 + y^2) \bmod 4 = 0$

then y

 $2 + x$
 $2 \bmod 4 = 0$.

R is also transitive because R contains all those pairs which consists of even numbers.

 $R = \{(2, 0), (2, 4), (4, 8), (8, 12) \dots\}$

So, if x

 $2 + y$
 $2 \bmod 4 = 0$ and y
 $2 + z$
 $2 \bmod 4 = 0$, then x
 $2 + z$
 $2 \bmod 4 = 0$.

So, R is transitive.

Q4 Text Solution:

Square of any number can't be less than zero, so, it will contain only $\emptyset$. and $\emptyset$ other than reflexivity satisfies all other properties. Since R is not reflexive hence option (a) is not correct.

Q5 Text Solution:
 xRy if $(x < 2y + 1)$

Reflexive as

 $x < 2x + 1$ {always true}

Symmetric:

example

 $(1, 5) 1 < 2(5) + 1$ but $(5, 1) \notin R$ ($5 < 2(1) + 1$)

Therefore, Not symmetric,

For Transitivity:

Take example of $(5, 3) \notin R$ & $(3, 2) \notin R$ but $(5, 2)$

 $\notin R$

Therefore, Not transitive.

Q6 Text Solution:

The reflexive closure of $R = \{(1, 2), (2, 1), (2, 3), (4, 4), (3, 4), (1, 1), (2, 2), (3, 3)\}$

So, its cardinality = 8.

Q7 Text Solution:

The symmetric closure of $R = \{(1, 2), (2, 1), (2, 2), (2, 4), (4, 2), (2, 3), (1, 1), (3, 3), (3, 2)\}$.

So, its cardinality = 9.

Q8 Text Solution:

The transitive closure of $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (3, 2), (2, 3), (2, 4), (4, 2), (1, 3), (3, 1), (1, 4), (4, 1), (3, 4), (4, 3)\}$.

So, its cardinality = 16.

Q9 Text Solution:

$(1, 1) \in R$, but $2, 2 \notin R$. So, it is neither reflexive nor irreflexive. Also, it is not asymmetric.

If $xy = 1$ then $yx = 1$. So, it is symmetric.

Since $(2, 1/2) \in R$ and $(1/2, 2) \in R$ but $(2, 2) \notin R$.

So, it is neither transitive nor anti-symmetric.

Q10 Text Solution:

Given $\{(a, b) | a \neq b\}$ means all possible pairs except self pair basically $\{(a, b) | a \neq b\} \cup \{(a, b) | a = b\} = Z \times Z$ therefore, (d) is correct.


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